MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
INSPECTION & ASSET INTEGRITY DIVISION — NDT SERVICES

ULTRASONIC THICKNESS (UT) GAUGING & CORROSION RATE EVALUATION REPORT
REPORT NO: **MRPL-NDT-UT-2026-089** | DATE OF SURVEY: 28-FEB-2026 | STATUS: **ACTION REQUIRED**

1. ASSET DETAILS & SURVEY CONDITIONS

Asset Tag / ID:	Feed Booster Pump P-101A & Suction Piping 4"-HC-1012-A1A1H	Plant Unit:	CDU-II / NHT Feed Train
Material / Spec:	ASTM A106 Gr. B (Carbon Steel) / Schedule 40	Operating Temp:	65 °C / 1.8 bar (g)
Nominal Thickness:	6.02 mm (Schedule 40)	Min. Required (t_min):	3.20 mm (per ASME B31.3)
NDT Equipment:	Olympus 38DL-Plus Ultrasonic Gauge (Calib Due: Nov-2026)	Couplant Used:	High-Temp Silicone Gel

2. ULTRASONIC THICKNESS MEASUREMENT GRID

Location / Point	Description / Orientation	Nominal (mm)	Previous (mm)	Current (mm)	Loss (mm)	Status
P1	Pump Suction Reducer Top	6.02	5.10	4.85	-0.25	ACCEPTABLE
P2	Pump Suction Reducer Bottom (6 o'clock)	6.02	4.60	4.15	-0.45	MONITOR
P3	Elbow #1 Extrados (4"-HC-1012)	6.02	4.30	3.85	-0.45	MONITOR
P4	Elbow #1 Intrados	6.02	5.30	5.12	-0.18	ACCEPTABLE
P5 (CRITICAL)	**Elbow #2 Extrados Impingement Zone**	**6.02**	**3.80**	**3.42**	**-0.38**	**NEAR RETIREMENT**
P6	Pump Volute Casing Inlet Throat	8.50	7.10	6.68	-0.42	ACCEPTABLE
P7	Bypass Tie-in T-piece (2"-HC-1015)	5.54	4.20	3.90	-0.30	MONITOR

3. REMAINING LIFE & CORROSION RATE CALCULATION

- **Calculated Corrosion Rate (CR) at Point P5**: (3.80 mm - 3.42 mm) / 1.35 yrs = **0.281 mm/year**.
- **Remaining Life (RL)**: (t_actual - t_min) / CR = (3.42 mm - 3.20 mm) / 0.281 mm/yr = **0.78 years (9.4 months)**.
- **ENGINEERING RECOMMENDATION**: Spool section containing Elbow #2 MUST be scheduled for replacement during upcoming CDU-II mini-shutdown (May 2026). Continuous operational monitoring required. Derate hydrotest pressure to 16.5 bar.

Inspected By: S. V. Shenoy (ASNT Level-II)
[Signed 28/02/2026]

Verified By: R. K. Sharma (Sr. Mgr NDT)
[Signed 01/03/2026]

Approved: Dy. GM (Inspection)
[STAMPED & APPROVED]